

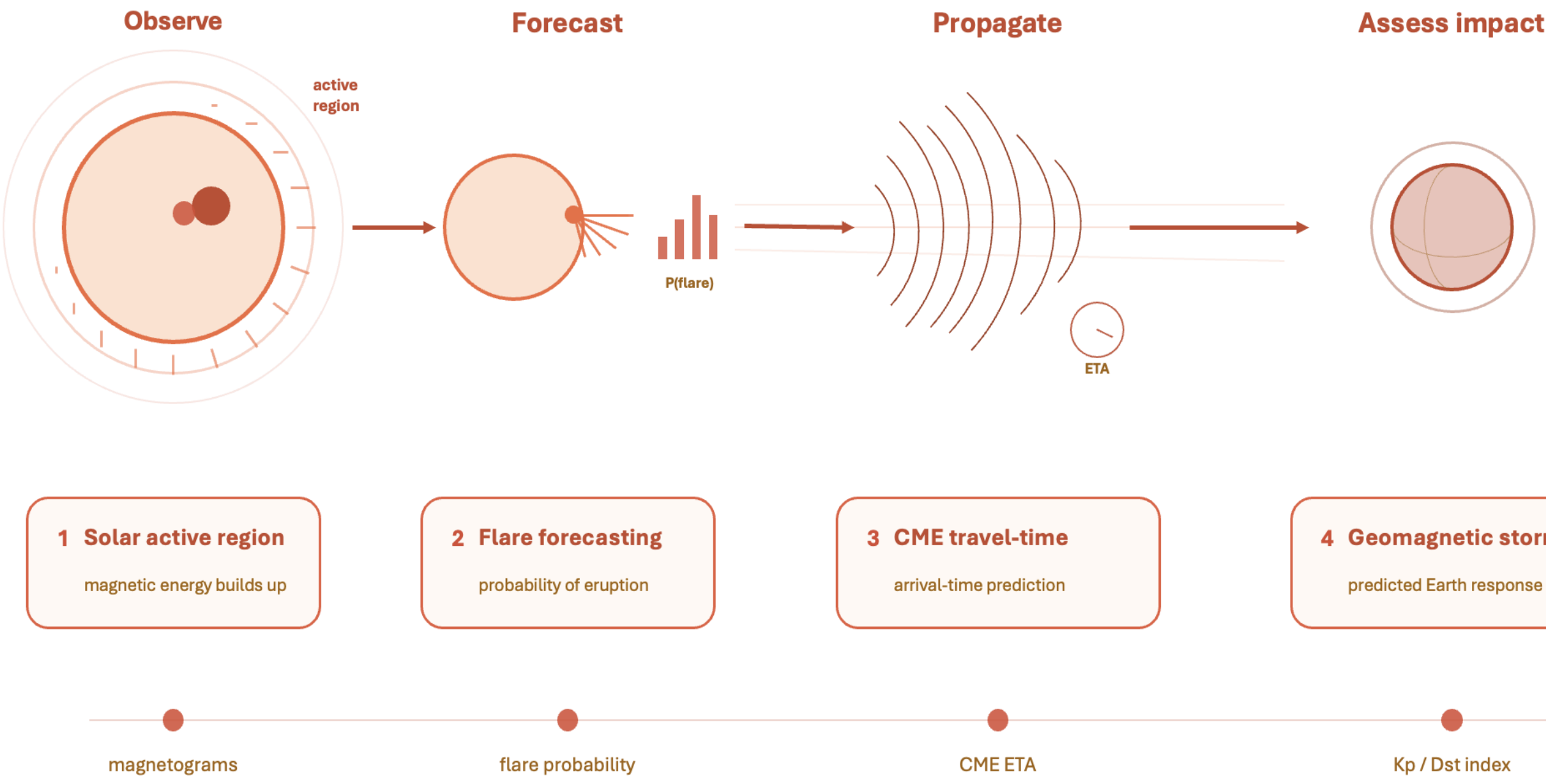
Physics-Driven Machine Learning for Operational Space Weather Forecasting

Edoardo Legnaro, Sabrina Guastavino, Anna Maria Massone, Michele Piana

University of Genova, MIDA, Department of Mathematics



UNIVERSITÀ
DEGLI STUDI
DI GENOVA



Coronal Mass Ejections (CMEs) consist of large burst of plasma and magnetic field that erupts from the Sun and travels through space, and they are typically triggered by solar flares.

To forecast the travel time, we start from the **Drag Based Model**, which describes the evolution of a CME as a function of the solar wind conditions

$$\ddot{r}(t) = -\gamma|\dot{r}(t) - w|(\dot{r}(t) - w)$$

$$\text{Drag Parameter } \gamma = C \frac{A\rho}{m}$$

ρ = solar wind density
 A = CME impact area
 m = CME mass
 C = drag coefficient

$\ddot{r}$ = CME acceleration
 $\dot{r}$ = CME speed
 w = solar wind speed

Hybrid Loss

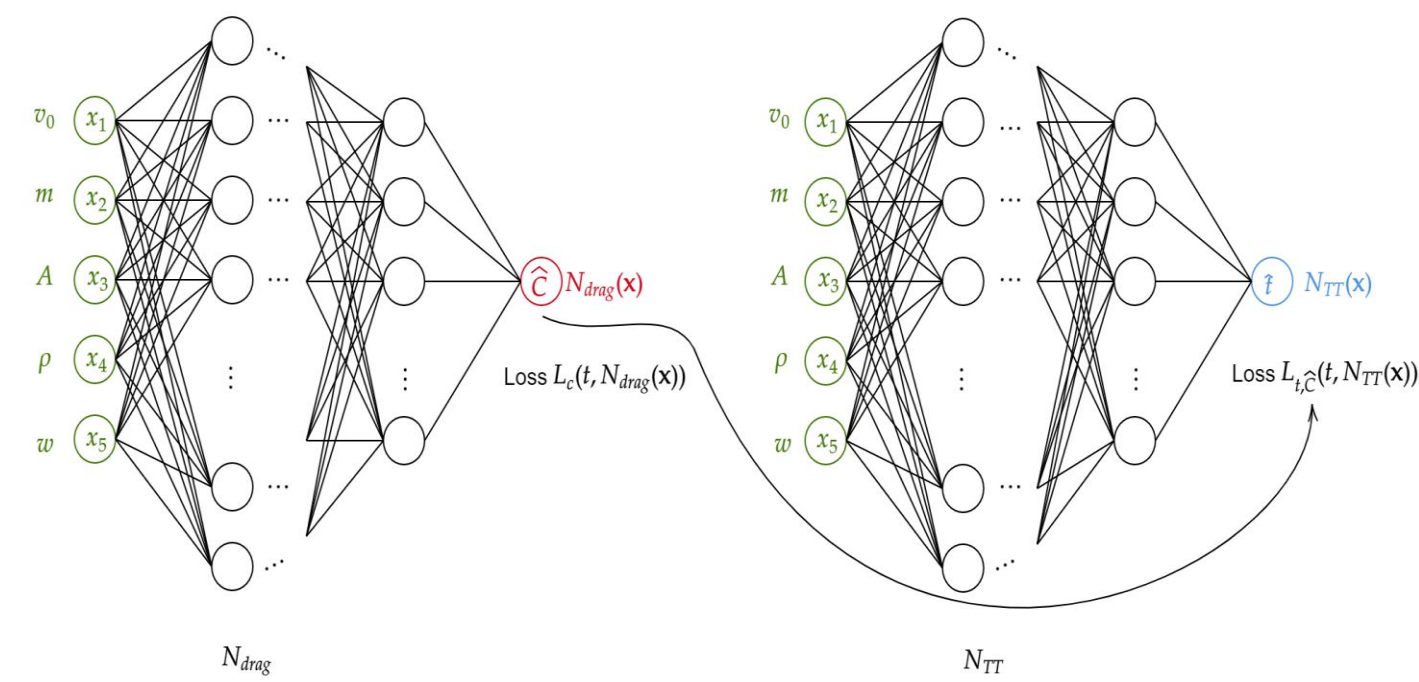
We include the drag-based model in the loss function
 λ is a regularization parameter

$$L_t(t, f(w, x)) = \lambda \left[(t - f(w, x))^2 + \text{Data-driven term (MSE)} \right] + (1 - \lambda) \left[1 - r(f(w, x), C) \right]^2 \text{ Physics-driven term}$$

The loss than has **two** different **components**

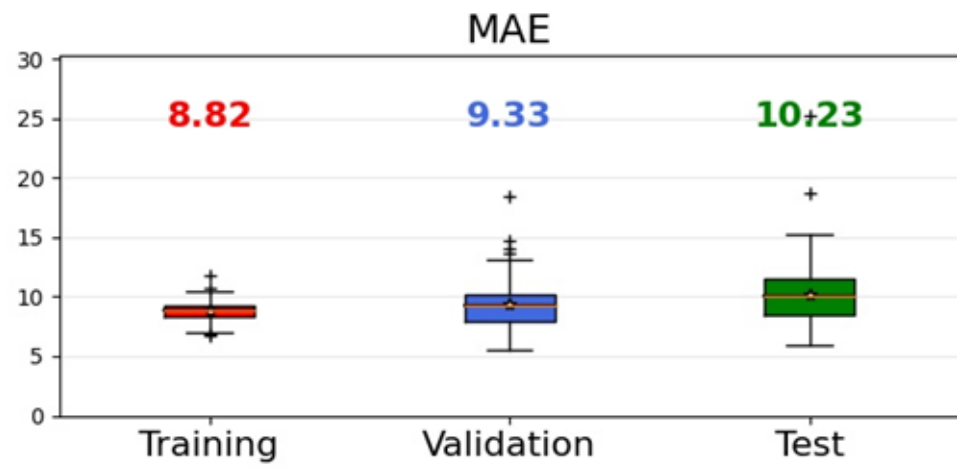
- **standard data-driven term**, which minimizes the error between the predicted and the true travel time.
- **physics driven term**: enforces that the predicted travel time is consistent with the drag-based equation

Physics-constrained architecture

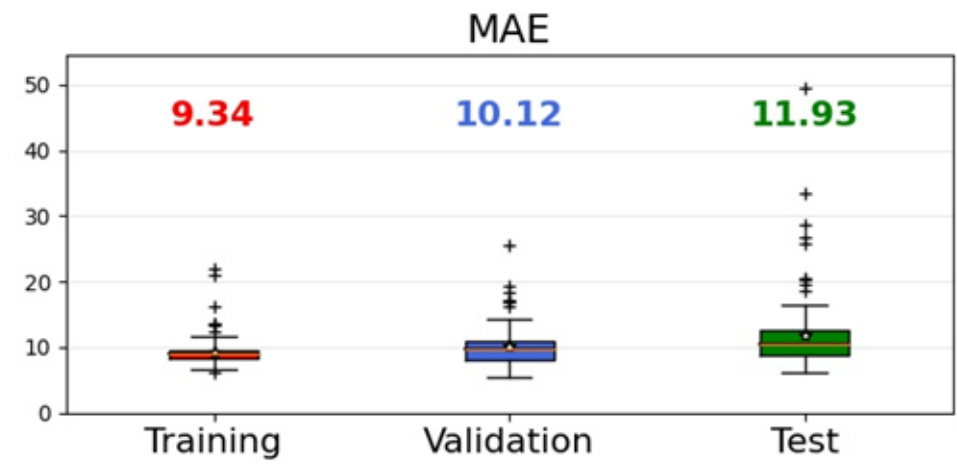


The first network estimates the drag coefficient. Its estimate constrains the travel-time network through the training loss, but it is not used as an inference input.

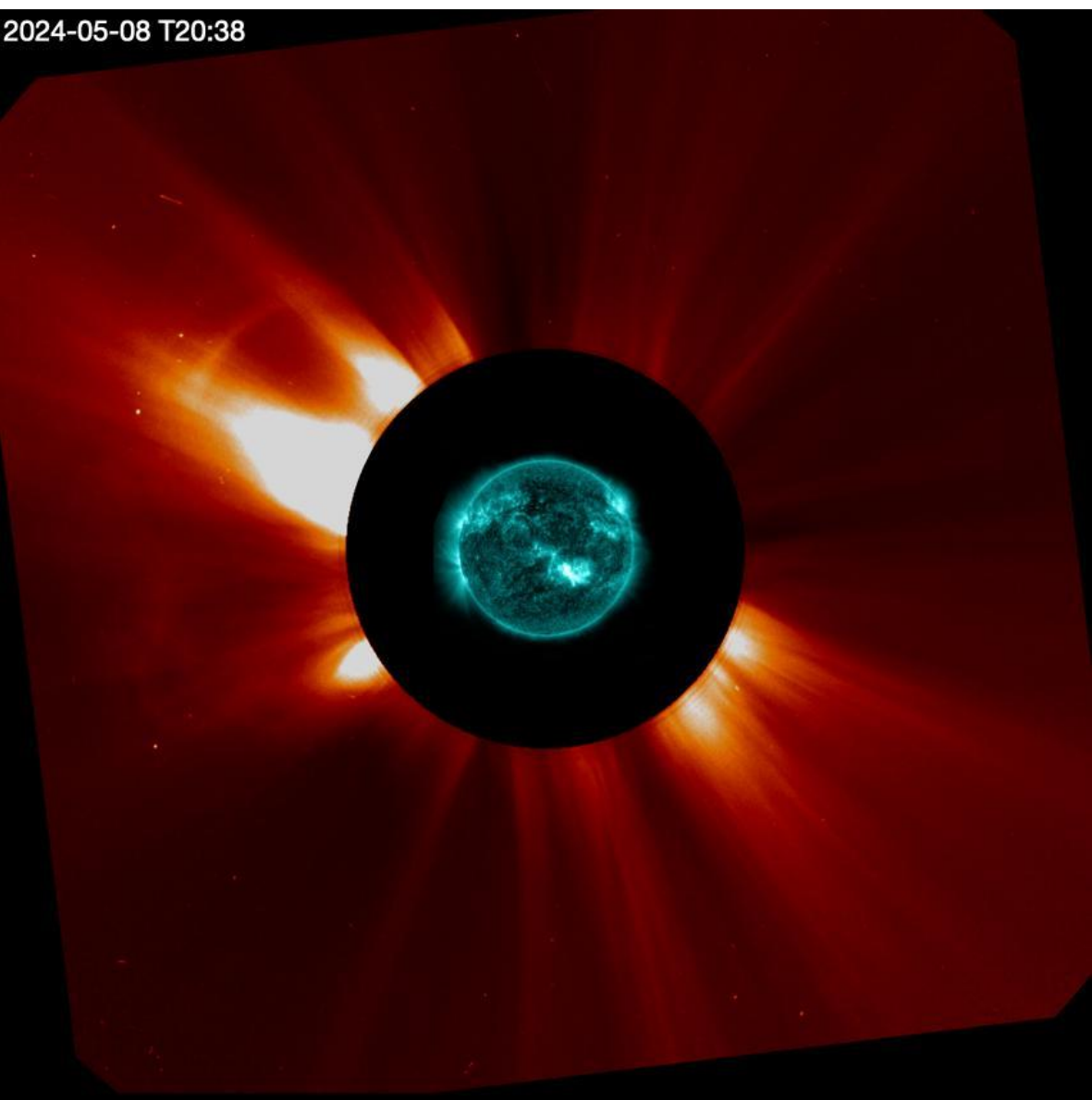
Physics Guided Net



Fully Data Driven Net



May 2024 superstorm case study



Observed CME inputs

v_0 2598.65 km/s m 1.37×10^{16} g
 ϕ 166.54° ρ 3 cm⁻³ w 500 km/s

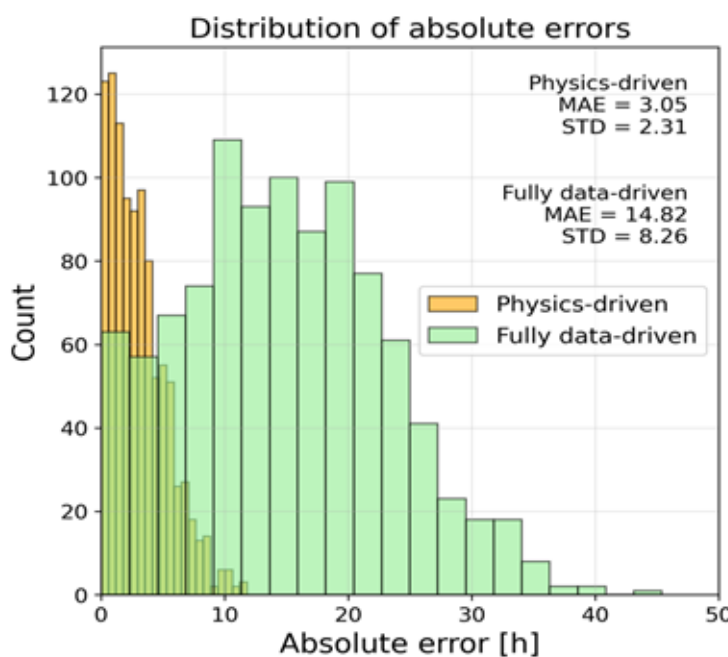
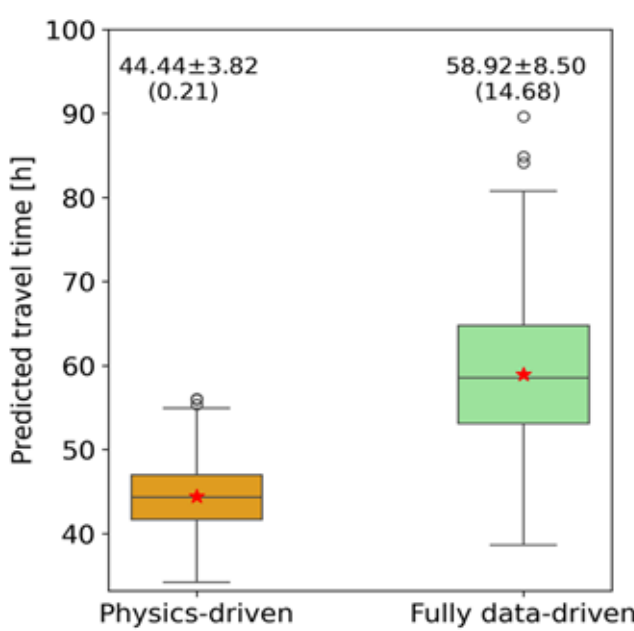
44.21 ± 3.20 h

predicted travel time

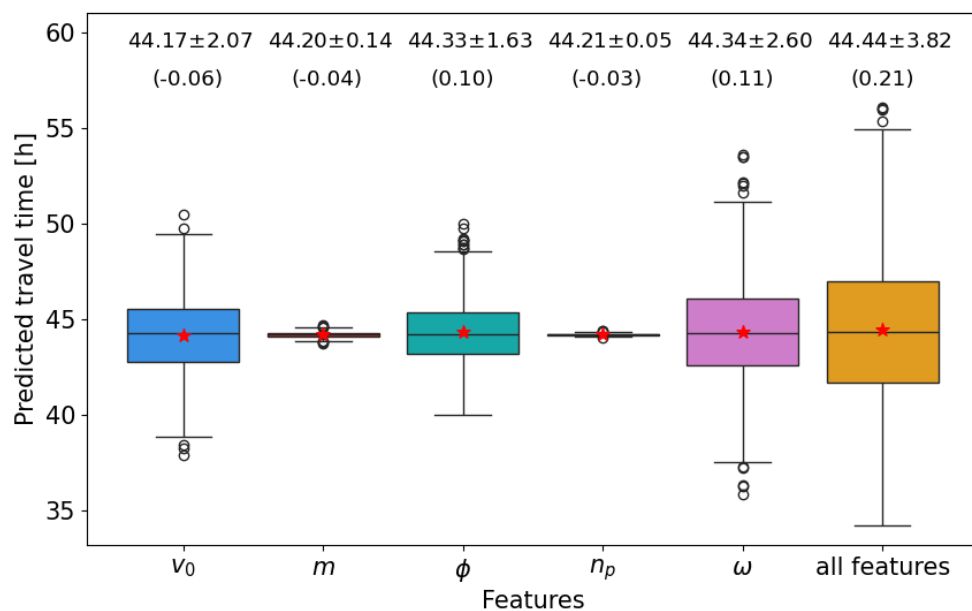
±3.20 h is the ensemble spread and reflects training-related epistemic uncertainty.

1 minute

arrival-time error (−0.02 h)



Robustness to input uncertainty



1000 realizations with Gaussian perturbations ($\sigma = 10\%$)

Initial CME speed v_0 and solar-wind speed w produce the largest spread. When all inputs vary, the mean prediction is 44.44 ± 3.82 h and the MAE is about 3 h.

Input uncertainty, distinct from the ±3.20 h ensemble spread

